

## 09.06 Trigonometric Functions of Any Angle

Common problems and solution methods

### Rule guide: recognize the task first

| Recognize                                                    | Apply these rules                                                                                              |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Trigonometric Functions And Models                           | Identify the angle measure, quadrant, or graph feature before selecting a formula.                             |
| The expression contains the chapter's key notation or model. | Write the known quantities, choose the matching formula, calculate carefully, and check the result in context. |

### Example 1: Identify the model

#### Problem

$$\sin 30^\circ = \frac{1}{2}$$

#### How to solve it

Write the known quantities and restrictions.

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

Check the result against the original statement.

### Example 2: Apply the key rule

#### Problem

$$135^\circ \cdot \frac{\pi}{180}$$

#### How to solve it

Write the known quantities and restrictions.

$$B = \frac{2\pi}{T}$$

Check the result against the original statement.

### Example 3: Check the result

#### Problem

$$y = 3\sin\left(\frac{\pi x}{2}\right) + 4$$

#### How to solve it

Write the known quantities and restrictions.

$$y = A\sin(Bx) + D$$

Check the result against the original statement.